ACRME Requirements Code Glossary & Master Index

Azure Capacity Reservation Management Engine (ACRME)
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Document Purpose

This glossary is the **single lookup point** for every requirements code used across the ACRME design corpus.
Each entry provides:

- The **code** and its full name
- A **one-line definition** of what it means
- A **direct link** to the source document and section where the full normative detail lives

It does not reproduce the full specification — it tells you exactly where to find it.

Code Family Quick Reference

Family	Prefix	Count	What it covers	Primary source
Region strategy	`REG-`	4	Region catalogue, distribution model, geography classification	Requirements Baseline §6
Strategic decisions	`DEC-`	3	Mandatory business/legal decisions that gate design	Requirements Baseline §23
Environment policy	`ENV-`	7	Prod / CVAL / DR environment separation rules	Requirements Baseline §7
Capacity reservation	`CAP-`	25	Reservation lifecycle, sharing, scaling, naming, zone structure	Requirements Baseline §8
Quota management	`QUA-`	14	Quota pooling, allocation, reclamation, governance	Requirements Baseline §9
Combined readiness	`RDY-`	4	Deployment readiness gate and state model	Requirements Baseline §10
Placement	`PLC-`	12	Customer placement, seed record, scoring, zone distribution	Requirements Baseline §11
Disaster recovery	`DR-`	19	DR sizing, failover, failback, co-location, standby activation	Requirements Baseline §12
FinOps & cost	`FIN-`	8	Idle cost, buffer economics, shared-DR overcommit accounting	Requirements Baseline §13

Family	Prefix	Count	What it covers	Primary source
AEP integration	`INT-`	7	Provisioning contract, idempotency, output spec	Requirements Baseline §14
State & data	`DAT-`	6	State store, entity model, freshness, config versioning	Requirements Baseline §15
Observability	`OBS-`	5	Metrics, alerts, dashboards, SLO reporting	Requirements Baseline §16
Governance & security	`GOV-`	9	RBAC, audit, break-glass, secrets, data classification	Requirements Baseline §17
Non-functional	`NFR-`	10	Availability, consistency, performance, idempotency	Requirements Baseline §18
Operational	`OPS-`	6	Runbooks, safe pause, override, naming convention	Requirements Baseline §19
Hard constraints	`HC-`	11	Hard blocking rules that gate every placement decision	Hard Constraints Reference
Validation rules	`VR-`	12	Named enforcement checks derived from hard constraints	Hard Constraints Reference
Core formulas	`A.`	9	Calculation formulas for reservations, quota, DR, zone distribution	Requirements Baseline Appendix A
	`C-`	13	Values and policies that are	

Family	Prefix	Count	What it covers	Primary source
Configurable items			configurable (not fixed)	Requirements Baseline §22
Pending decisions / POCs	`POC-` / `DEP-`	12	Azure behaviour that must be validated before production dependency	Requirements Baseline §23
Functional requirements	`FR-`	8	Top-level functional capability areas	Complete Requirements Reference
Non-functional requirements (placement)	`NFR-R`	3	Latency, freshness, determinism for placement API	Complete Requirements Reference
Placement requirements	`R`	8	Specific placement behaviours derived from placement design	Complete Requirements Reference
Architecture decisions	`ADR-`	5	Formal load-bearing architectural decisions	Architecture/adr/
Design gaps	`G-`	4	Explicitly documented open design items (30% gap)	UML Class Diagrams Summary
Backlog epics	`ACRME-E`	20	Top-level delivery epics in the engineering backlog	Backlog Epics/Stories/Tasks

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REG — Region Strategy Requirements

Primary source: Requirements Baseline v2.4 — §6 Region Strategy & Classification

Code	Name	One-line definition
REG-001	Configurable region catalogue	Region classification, eligibility, distribution model, and per-region flags are all configuration-driven and versioned; no hard-coded catalogue in engine code.
REG-002	Cross-geo DR extension region	Switzerland North is the authoritative cross-geography DR extension region (Europe geography); cross-geo DR path requires explicit policy approval (HC-10).
REG-003	Distribution model	US uses a three-region distribution model (Prod / CVAL / DR fully separated); all other geographies (Europe, Australia, Asia Pacific, Middle East) use a two-region model.
REG-005	Japan East pending	Japan East is pending business confirmation before inclusion as a Standard region in the Asia Pacific geography.

DEC — Strategic Decisions

Primary source: Requirements Baseline v2.4 — §23 Pending Decisions & Mandatory POCs

Code	Name	One-line definition
DEC-001	Middle East DR offering	Legal/business decision on Middle East DR (`DR_NOT_OFFERED` is the current default until this decision records an approval per country/geography). Highest-priority gating decision.
DEC-002	DR drill duration & failback	Choice between extended ~1-year DR run and earlier ~30-day failback model.
DEC-003	Geography exception approval	Who approves geography-only customer onboarding (exception path) and the format of binding customer acknowledgement.

ENV — Environment Policy Requirements

Primary source: Requirements Baseline v2.4 — §7 Environment Policy Requirements

Code	Name	One-line definition
ENV-001	Production reservation coverage	Manage approved production VM SKUs with reservations to guarantee compute availability.
ENV-002	Production-only initial enforcement	Production reservation coverage is mandatory; CVAL/DR reservation is a phased follow-on, not day-one scope.
ENV-003	Hard separation constraints	Prod, CVAL, and DR must each be in a different region (US three-region); in two-region geographies CVAL and DR co-locate in the non-Prod region (PLC-010a — mandatory, not an exception).
ENV-004	CVAL treatment	CVAL is treated as a potential DR capacity source (its capacity can contribute to DR readiness).
ENV-005	DR bootstrap, not full duplicate	DR capacity is a lean bootstrap (enough to start recovery orchestration), not a full production duplicate.
ENV-006	DR bootstrap cannot be implicitly zero	A zero DR target must be an explicit approved configuration, not an accidental default.
ENV-007	R&D policy	No permanent R&D reservations by default; short-lived POC reservations may be supported under governance.

CAP — Capacity Reservation Management Requirements

Primary source: Requirements Baseline v2.4 — §8 Capacity Reservation Management Requirements

Code	Name	One-line definition
CAP-001	Authoritative managed scope	The engine operates only on subscriptions, regions, SKUs, and environments declared in its versioned scope file.
CAP-001a	Core subscription = all production	VMs deployed into a shared platform/core subscription are all classified production for reservation and buffer purposes; non-production workloads must not run there.
CAP-002	Azure resource must precede config activation	A managed subscription/CRG must exist as a real Azure resource before the engine activates management of it.
CAP-003	Reservation target formula	Reserved quantity = allocated demand + approved buffer; see formula A.1.
CAP-004	Associated-but-deallocated VMs	Deallocated VMs associated to a CRG still consume reserved capacity; they must not be silently ignored in demand accounting.
CAP-005	Automated reconciliation	Periodically compare reserved quantity against current demand; trigger scale-up or flag scale-down.
CAP-006	Reconciliation frequency (configurable)	Reference implementation reconciles every 6 minutes; production interval is configurable (C-4) and validated by POC-010.
CAP-007	Scale-up behaviour	When allocated demand rises, attempt to increase reservation immediately; log failure with alert if Azure rejects.
CAP-008	Scale-down behaviour	When allocated demand falls, reduce reservation only after the decommissioning work-

Code	Name	One-line definition
		flow completes; not automatic.
CAP-009	Zero-capacity support	Where Azure permits, reduce a reservation to `quantity = 0` (not delete it) to preserve the CRG structure; extends CAP-022 seed matrix.
CAP-010	No automatic deletion by default	CRGs and reservations are never automatically deleted; decommissioning is a gated workflow (CAP-008).
CAP-011	Availability-zone isolation	Reservations are managed per availability zone; zone counts and zone-level capacity are tracked and reported independently (extends to CAP-023 per-AZ CRG structure).
CAP-012	Regional isolation	Reservations are never counted, moved, or shared across regions without explicit policy.
CAP-013	Capacity sharing	Support cross-subscription reservation sharing where approved; governed by the three-tier sharing model (Tier 1/2/3).
CAP-014	Shared capacity visibility	For each shared reservation, expose provider-side quantity, consumer-side allocated count, and net available capacity.
CAP-015	No double counting	Capacity shared to multiple consumers is counted once at the provider; consumer-allocated demand is tracked separately.
CAP-016		Validate capacity availability before accepting a deploy-

Code	Name	One-line definition
	Reservation integrity validation (pre-deploy)	ment request; block if insufficient.
CAP-017	Deployment failure policy	If reservation enforcement fails at deploy time, the engine follows a defined policy (block / warn / allow with alert) — configurable per environment.
CAP-018	Over-allocation policy	Support an explicit policy to permit VM association beyond reserved quantity (with alert); not silent.
CAP-019	Scope-file governance	Adding a SKU to management requires a scope-file change with approval; reconciles with CAP-024 reactive discovery.
CAP-020	Availability-Set VMs ineligible	VMs in an Availability Set are mutually exclusive with Capacity Reservations and must be excluded from all reservation management (maps to HC-11).
CAP-021	Deallocate-or-migrate-to-AZ onboarding precondition	Before onboarding an AV-Set VM, it must be deallocated and redeployed into an availability zone — this is a precondition, not an engine action.
CAP-022	Seed-at-0 eligible-SKU/AZ matrix & budget governance	Maintain a governed matrix of eligible SKU × region × AZ combinations with seed reservations created at `quantity = 0`; resizing from 0 requires product-team budget approval. Extends CAP-009.
CAP-023	Regional and per-AZ CRG structure	For each environment, the engine maintains one regional CRG (non-zonal SKUs) plus one per-AZ CRG for each

Code	Name	One-line definition
		availability zone (`crg-<env>-<region>-az1/az2/az3`). Extends CAP-011.
CAP-024	Reactive SKU/AZ discovery reconciled with governance	Reactive discovery auto-creates CRGs for new SKU/AZ combinations observed in demand; governed by the scope-file approval workflow (CAP-019).

QUA — Quota Management Requirements

Primary source: Requirements Baseline v2.4 — §9 Quota Management Requirements

Code	Name	One-line definition
QUA-001	Separate quota domain	Manage quota as a separate control plane from reservations; quota failure does not silently mask reservation success.
QUA-002	Regional & family scope	Maintain quota inventory by region and VM family for every managed subscription.
QUA-003	Central pooling ("quota hoarding")	Where Azure quota is per-subscription by default, centralise it into a governed pool for controlled reallocation (quota hoarding governance practice).
QUA-004	Single governed pool per supported scope	Prefer one governed quota pool per region/VM-family scope rather than fragmented per-subscription pools.
QUA-005	Quota as cost/consumption governor	Quota allocation is a cost-control gate, not just a technical limit; changes require business justification.
QUA-006	Production growth buffer	Production subscriptions hold a quota buffer above current usage to absorb growth without an increase request per deployment.
QUA-007	Quota–reservation validation	For every managed scope, quota available ≥ reserved quantity (or alert); a reservation without supporting quota is flagged `READY_WITH_RISK`.
QUA-008	Dynamic allocation	Allocate quota from the pool to a subscription on demand; reclaim when no longer needed.
QUA-009	Quota reclamation	Reclaim unused subscription quota into the pool on scope reduction, subscription off-

Code	Name	One-line definition
		boarding, or environment retirement.
QUA-010	Quota request governance	Increase requests record reason, approver, business justification, and cost exposure before submission to Azure.
QUA-011	Quota source discovery	Discover reusable quota assigned to decommissioned or lightly used subscriptions before requesting new quota from Azure.
QUA-012	Region-failure assumption	Quota bound to a failed region cannot be assumed available; failover planning must not depend on failed-region quota.
QUA-013	Consumer-subscription quota (POC-gated)	Assume a VM deploying against a shared reservation still requires quota in the consumer subscription — but this behaviour must be confirmed by POC-001 before production dependency.
QUA-014	Quota allocation audit	Log every allocation, reclamation, increase request, and approval with timestamp, actor, and justification.

RDY — Combined Capacity & Quota Readiness

Primary source: Requirements Baseline v2.4 — §10 Combined Capacity & Quota Readiness

Code	Name	One-line definition
RDY-001	Deployment readiness gate	A deployment is capacity-ready only when reservation exists, quota is sufficient, and state is fresh — all three must be satisfied.
RDY-002	Readiness states	The engine exposes a machine-readable readiness state: `READY` , `READY_WITH_RISK` , `QUOTA_DEFICIT` , `CAPACITY_DEFICIT` , `STALE_STATE` , `NOT_MANAGED` .
RDY-003	Capacity and quota must agree	A reservation without sufficient deployable consumer quota is never `READY` ; it is at minimum `READY_WITH_RISK` or `QUOTA_DEFICIT` .
RDY-004	No stale placement	Region selection and provisioning must not use state older than the configured maximum freshness threshold; stale state fails safe (`STALE_STATE`).

PLC — Region Selection & Customer Placement Requirements

Primary source: Requirements Baseline v2.4 — §11 Region Selection & Customer Placement

Code	Name	One-line definition
PLC-001	Production region is the primary input	The default onboarding path starts with the customer supplying an exact production region; geography-only is an exception requiring approval.
PLC-002	Geography-based selection is exceptional	If a customer supplies only a geography (not an exact region), the engine selects via `argmax(PS_Prod)` but requires explicit exception approval (DEC-003).
PLC-003	Customer seed record	The first placement decision creates an authoritative seed record holding Prod, CVAL, and DR regions; subsequent placements reuse it.
PLC-004	Reuse across products	All products/environments for the same customer reuse the same seed record regions; no per-product re-selection.
PLC-005	Controlled seed change	The seed record is never regenerated automatically; changes require an explicit operator action with reason and approval.
PLC-006	CVAL & DR selection	Once production is fixed, the engine selects CVAL and DR regions automatically, subject to hard constraints and separation rules.
PLC-007	Live weighting	Placement scoring considers live allocated demand (demand units, not customer count) and capacity headroom to prevent hotspots.
PLC-008	Lowest suitable load	Select the lowest-risk suitable region; avoid regions at or near capacity ceiling.

Code	Name	One-line definition
PLC-009	AEP-triggered pipeline	Region selection is the first step in every AEP provisioning call; the pipeline is synchronous for the recommendation, async for the reservation action.
PLC-010	CVAL/DR co-location	A customer's CVAL and DR may share a region — this is the normal outcome in two-region geographies, not an exception.
PLC-010a	Co-location mandatory in two-region geographies	In any two-region geography, CVAL and DR **must** co-locate in the non-Prod region — there is no third region available.
PLC-011	Even zone-distribution target & rebalancing	Maintain an approximately even `≈ 1/zone_count` zone distribution for all placements; trigger rebalancing when skew exceeds the C-13 tolerance (formula A.9).

DR — Disaster Recovery Capacity Requirements

Primary source: Requirements Baseline v2.4 — §12 Disaster Recovery Capacity Requirements

Code	Name	One-line definition
DR-001	Single-region failure basis	The default model plans for one region failing at a time; simultaneous multi-region failure is out of scope.
DR-002	Distributed DR	DR capacity is computed from the distributed multi-source, multi-destination reference model (Section 12A).
DR-003	Destination distribution	Record how each source region's workload is distributed across destination regions; maintain the source→destination DR index (DR-018).
DR-004	CVAL target contribution	Normal CVAL placement should land in the same region as the planned DR destination so CVAL capacity earmarks DR bootstrap slots.
DR-005	CVAL may exceed DR need	CVAL capacity may exceed the DR bootstrap requirement; the excess earmark is tracked but not wasted.
DR-006	Staged DR capacity acquisition	DR capacity is acquired in waves (CVAL earmark → bootstrap top-up → full demand) rather than reserved all-at-once.
DR-007	DR target is configurable	Bootstrap quantity may be a node count, core count, percentage of source, or formula result; all are configurable (C-1).
DR-008	Control plane first	Bootstrap prioritises the required control-plane nodes and SKUs needed to run recovery orchestration before general workload capacity.

Code	Name	One-line definition
DR-009	Recovery prioritisation	Support prioritised recovery: critical workloads recover first, lower-priority workloads recover in sequence.
DR-010	DR declaration guardrail	Destructive/service-impacting failover actions require an explicit authorised DR declaration; no silent auto-failover.
DR-011	Source region not a capacity source during outage	During a declared outage, the failed source region's capacity is unavailable and must not be counted in failover planning.
DR-012	DR drill rotation	Support periodic DR drills with role-flip (source becomes destination and vice versa) and role restoration after the drill.
DR-013	Failback policy (configurable)	Support extended-run failback (~1 year) and shorter failback (~30 days); configurable per customer/geography (C-5, DEC-002).
DR-014	Middle East policy flag	Support `DR_NOT_OFFERED` per geography; when set, no DR region is assigned and cross-geo DR substitution does not apply. See DEC-001.
DR-015	Future active-active consideration	Note that active-active DR is out of current baseline scope; architecture should not block its future addition.
DR-016	Reciprocal multi-source hosting	Every region may simultaneously be a source (hosting live workloads) and a destination (hosting standby for another region).
DR-017		Because DR-001 assumes only one region fails at a

Code	Name	One-line definition
	Non-concurrent capacity sharing (max, not sum)	time, a destination's DR capacity requirement is the **maximum** over its non-concurrent sources — not the sum (Appendix D).
DR-018	Source→destination DR index	Maintain an authoritative index recording which source regions each destination protects and the failover portion size.
DR-019	Standby activation on declaration	On an authorised DR declaration, the engine activates the pre-positioned standby capacity in the destination region in priority waves.

FIN — Cost Economics & FinOps Requirements

Primary source: Requirements Baseline v2.4 — §13 Cost Economics & FinOps Requirements

Code	Name	One-line definition
FIN-001	Idle cost measurement	Calculate and surface the cost of unused reserved capacity (idle reservations) per scope.
FIN-002	Configurable economic policy	Buffer and bootstrap values are configurable so cost exposure can be tuned without code changes.
FIN-003	Cost before expansion	Before increasing a persistent reservation, expose the projected monthly cost increase so the operator can approve with cost awareness.
FIN-004	Cost allocation	Reservation cost is attributable to the owning team/product/environment for chargeback.
FIN-005	Underutilisation review	Long-running underutilised reservations trigger a review notification rather than automatic deletion.
FIN-006	No cost-only unsafe reduction	Cost optimisation never overrides a safety floor (HC-2, HC-6, HC-7); cost savings are bounded by capacity commitments.
FIN-007	Audit-friendly flexibility	Buffer/bootstrap values are versioned and traceable; historical cost decisions are explainable.
FIN-008	Shared-DR overcommit accounting	Where DR-017 max-not-sum sizing overcommits a destination, the overcommit ratio is calculated and exposed per destination for leadership sign-off.

INT — Provisioning & AEP Integration Requirements

Primary source: Requirements Baseline v2.4 — §14 Provisioning & AEP Integration

Code	Name	One-line definition
INT-001	Capacity API/workflow	AEP calls a capacity+placement API or triggers a workflow; ACRME returns a readiness verdict and placement recommendation.
INT-002	Idempotent interface	Capacity checks and reservation actions are idempotent; AEP may retry without side effects.
INT-003	Required input	The AEP call must supply: customer/realm; product; environment; exact region or geography; SKU/VM family; intended demand.
INT-004	Required output	ACRME returns: resolved Prod/CVAL/DR placement; readiness state; capacity available; quota available; operation handle for async tracking.
INT-005	Deployment reservation association	When policy requires it, ACRME associates the newly deployed VM to the reservation after AEP confirms deployment.
INT-006	Concurrent deployment control	Prevent two concurrent AEP calls from double-committing the same capacity; use optimistic locking or a deployment lock.
INT-007	Partial-failure handling	If quota is allocated but reservation fails, the engine compensates (rolls back the quota allocation) and returns a structured error.

DAT — State & Data Requirements

Primary source: Requirements Baseline v2.4 — §15 State & Data Requirements

Code	Name	One-line definition
DAT-001	Authoritative state store	Maintain an authoritative desired-state store (Cosmos DB) as the single source of truth; Azure is the observed-state source.
DAT-002	Minimum entities	The state store must hold: subscriptions; regions/zones; SKUs/quota families; CRGs/reservations; seed records; assignments; snapshots; operation records; audit trail.
DAT-003	Freshness metadata	Every observation record carries a collection timestamp and a staleness flag; stale records block placement (RDY-004).
DAT-004	Historic state	Retain history sufficient to explain any past placement or sizing decision; minimum retention configurable.
DAT-005	Configuration versioning	Scope files and policies are versioned; every decision record references the policy version in force at decision time.
DAT-006	Schema compatibility	API and config schema changes follow a compatibility policy (additive changes only; breaking changes require version bump).

OBS — Observability & Alerting Requirements

Primary source: Requirements Baseline v2.4 — §16 Observability & Alerting

Code	Name	One-line definition
OBS-001	Core metrics	Emit: reserved quantity; allocated VM count; associated VM count; quota available; DR floor headroom; idle cost; reconciliation latency; staleness age.
OBS-002	Required alerts	Alert on: production buffer below target; DR capacity below floor; quota exhaustion; stale state; deployment blocked; unauthorised DR consumption; failed reconciliation.
OBS-003	Alert context	Every alert includes: region; zone; subscription; SKU/family; policy version; last-known-good timestamp; remediation link.
OBS-004	Dashboards	Provide dashboards at: regional; product; subscription; SKU; and DR-pair granularity.
OBS-005	SLO reporting	Expose SLO metrics: availability of the placement decision path; reconciliation latency P99; data freshness percentage.

GOV — Governance, Security & Compliance Requirements

Primary source: Requirements Baseline v2.4 — §17 Governance, Security & Compliance

Code	Name	One-line definition
GOV-001	Least privilege	Workload identities receive only the minimum permissions required; no standing owner/contributor on managed subscriptions.
GOV-002	Separation of duties	Policy approval, production mutations, and audit-log access are held by different roles; no single identity can approve and execute a high-impact change.
GOV-003	Scoped automation	Automation is constrained by tenant, management group, and subscription scope; no accidental cross-tenant or cross-geography blast radius.
GOV-004	Change approval	High-impact changes (reducing production reservations, deleting a CRG, emergency override) require a named approver and a tracked approval record.
GOV-005	Break-glass controls	Break-glass access requires an authorised request with time limit, reason, and post-action audit review.
GOV-006	Audit evidence	Every decision and mutation writes an immutable audit record with before/after values, actor, timestamp, and correlation ID.
GOV-007	Policy exceptions	Each exception (geography-only onboarding, over-allocation, cross-geo DR) has an owner, reason, expiry, and renewal process.
GOV-008	Secrets & credentials	No secrets in config files or logs; all credentials managed

Code	Name	One-line definition
		via Azure Key Vault and Managed Identity.
GOV-009	Data classification	Customer placement and workload metadata is classified and handled per organisational data-classification policy.

NFR — Reliability & Non-Functional Requirements

Primary source: Requirements Baseline v2.4 — §18 Reliability & Non-Functional Requirements

> Note: The complete requirements reference uses `NFR-1..NFR-7` for the top-level functional NFR areas and `NFR-001..NFR-010` for the baseline detailed items. Both refer to the same concepts at different granularities.

Code	Name	One-line definition
NFR-001	Availability	The capacity decision path is not an uncontrolled single point of failure; degraded-mode behaviour is defined and tested.
NFR-002	Consistency	Concurrency control prevents double-committing capacity or quota; optimistic locking on all state-changing operations.
NFR-003	Performance	Readiness responses meet an approved latency target; long-running Azure operations return a tracked async handle rather than blocking the caller.
NFR-004	Scale	Operates across hundreds of subscriptions, multiple regions/zones, VM families, products, and seed records without degradation.
NFR-005	API-throttling resilience	Use batching, caching, exponential backoff with jitter, retry limits, and per-scope rate control; avoid Azure API storms during a regional event.
NFR-006	Recoverability	State store, configuration, and audit trail support backup/restore and reconstruction of desired state after data-layer failure.
NFR-007	Idempotency	All mutating operations are idempotent or protected by a durable operation key; replayed calls have no additional effect.
NFR-008	Testability	Policies, formulas, reconciliation logic, weighting, failover

Code	Name	One-line definition
		distribution, and compensation are independently unit-testable without Azure dependency.
NFR-009	Simulation mode	Support read-only/dry-run that shows intended reservation/quota/placement/cost changes without applying them — including DR failover simulation.
NFR-010	Explainability	Every placement and sizing decision exposes the policy inputs, state snapshot, formula applied, and human-readable reason code.

OPS — Operational Requirements

Primary source: Requirements Baseline v2.4 — §19 Operational Requirements

Code	Name	One-line definition
OPS-001	Runbooks	Runbooks exist for: production buffer deficit; quota exhaustion; reservation scale-up failure; stale state; blocked deployment; DR declaration; CVAL shutdown; quota allocation to DR; reservation sharing activation/revocation; reconciliation rollback/pause; manual emergency override; regional recovery and failback; standby DR activation sequence (DR-019).
OPS-002	Safe pause	Operators can pause the engine’s mutation actions while retaining inventory collection and alerting (read-only mode).
OPS-003	Manual override	Authorised operators can set temporary desired values with an expiry timestamp and reason; the engine never overwrites an active approved override.
OPS-004	Maintenance-window awareness	Planned maintenance that changes VM allocation or association is visible to the engine to prevent inappropriate scaling triggers or alert noise.
OPS-005	Ownership	Every region, subscription, reservation scope, quota pool, alert, and exception has a named owning team and documented escalation route.
OPS-006	Naming convention & counter	All managed resource groups, CRGs, and subscriptions follow a deterministic, parseable naming convention. Pattern: RG `rg-odcr-<env>-<region>-<NN>` ; CRG `crg-<env>-<re-`

Code	Name	One-line definition
		`gion>-<scope>` (scope ∈ {reg, az1, az2, az3}); Subscription `sub-<org>-<domain>-<pur-pose>-<NN>`. See C-12 for configurability.

HC — Hard Constraints

Primary source: Hard Constraints Reference

Hard constraints are **blocking rules** applied by the placement pipeline before any scoring. A candidate region that fails any HC is eliminated — scoring is not applied to it.

Code	Name	One-line definition
HC-1	REGION_SEPARATION	Prod, CVAL, and DR must be in distinct regions (US three-region model). In two-region geographies, CVAL and DR co-locate in the non-Prod region under PLC-010a.
HC-2	CAPACITY_FLOOR	Reserved capacity must not be reduced below the allocated VM count; the floor is `quantity ≥ allocated`.
HC-3	QUOTA_FLOOR	Available quota in the target scope must be ≥ the vCPU count of the proposed reservation or deployment.
HC-4	DR_SEPARATION_CLASS	DR regions must carry a different environment classification from Prod; a region designated Prod cannot simultaneously serve as DR.
HC-5	ZONE_AVAILABILITY	A target availability zone must exist and be usable in the target subscription/region before a zonal placement is accepted.
HC-6	DR_COVERAGE_FLOOR	NonProd/CVAL placement in a DR-designated region must not reduce available capacity below the region's DR floor (the bootstrap reserved for failover).
HC-7	DR_FLOOR_INTEGRITY	The engine must enforce the DR floor continuously; any reconciliation action that would bring available capacity below the floor is blocked.
HC-8	GEOGRAPHY_CONTAINMENT	A derived Prod region must fall within the Standard Capacity Regions for the custom-

Code	Name	One-line definition
		er's chosen geography (not in a different geography).
HC-9	STANDARD_REGION_ONLY	All automated placement paths (geography-based, Prod derivation, NonProd/DR selection) must use Standard Capacity Regions only; Restricted regions are exception-path only.
HC-10	CROSS_GEO_EXTENSION_PATH_APPROVED	A cross-geography DR extension (e.g., Middle East Prod → Switzerland North DR) must be explicitly approved in the active PlacementPolicy before it is used.
HC-11	AVAILABILITY_SET_INELIGIBLE	VMs in an Azure Availability Set cannot be associated with a Capacity Reservation; they must be excluded from all reservation management (CAP-020).

VR — Validation Rules

Primary source: Hard Constraints Reference — Validation Rules table

Validation rules are named enforcement checks that implement hard constraints in the placement pipeline code. They are the testable, auditable expression of the HC rules.

Code	Name	One-line definition	Enforces
VR-1	Prod ≠ DR region	Prod and DR must not share a region.	HC-1
VR-2	Prod ≠ CVAL region	CVAL and Prod must not share a region.	HC-1
VR-3	CVAL/DR co-location policy	CVAL and DR may share a region only under the approved PLC-010a co-location policy (two-region geographies).	HC-1
VR-4	Prod within chosen geography	The derived Prod region must be within the Standard Capacity Regions for the customer's chosen geography.	HC-8
VR-5	Automated paths use Standard regions only	All automated placement paths must use Standard Capacity Regions only.	HC-9
VR-6	CVAL/DR never in Restricted regions	CVAL and DR must not use Restricted Capacity Regions under any condition, including exception deployments.	HC-9
VR-7	Exhaustion handling	If all Standard regions for a geography are eliminated by HC-1..HC-10, the engine returns a capacity exhaustion error with an ops alert.	HC-1..HC-10
VR-8	Middle East DR default	For Middle East geography with `DR_NOT_OFFERED = true` (current legal position, DEC-001), the seed record **must**	HC-10

Code	Name	One-line definition	Enforces
		record `dr_region = NOT_OFFERED`; no auto-assignment of Switzerland North.	
VR-8a	Middle East conditional cross-geo	Only after a recorded DEC-001 approval clears `DR_NOT_OFFERED`: Switzerland North may be used as a cross-geo DR; if it then fails HC-1..HC-10, block with an ops alert (VR-9 applies).	HC-10
VR-9	Cross-geo fallback exhaustion	On the DEC-001-approved conditional path only: if Switzerland North fails HC-1..HC-10, block placement with an ops alert.	HC-1..HC-10
VR-10	Restricted region exception path	A Restricted region requested by a customer triggers the Scenario 2 exception path only; it is never auto-selected.	HC-9
VR-11	Cross-geo extension approval	A cross-geography DR extension must be explicitly approved in the active PlacementPolicy before it can be used.	HC-10

A. — Core Formulas (Appendix A)

Primary source: Requirements Baseline v2.4 — Appendix A — Core Formulas

Code	Name	One-line definition
A.1	Reservation target	`reserved_target = allocated + buffer` ; the approved buffer is configurable per product/region/env/SKU (C-2).
A.2	Reservation headroom	`headroom = reserved - allocated` ; positive headroom means spare committed capacity.
A.3	Reservation deficit	`deficit = allocated - reserved` ; a positive deficit means the reservation is under-sized for current demand.
A.4	Available subscription quota	`available_quota = quota_limit - quota_current_value` ; the raw Azure-reported available quota.
A.5	Deployment quota deficit	`quota_deficit = committed_vCPUs - available_quota` ; positive means the reservation cannot be backed by deployable quota.
A.6	DR destination requirement (CORRECTED)	`DR_req(dest) = MAX over non-concurrent sources s protected by dest ( portion(s → dest) )` ; max-not-sum because DR-001 assumes only one source fails at a time (Appendix D).
A.7	DR capacity gap	`dr_gap(dest) = DR_req(dest) - dr_reserved(dest)` ; positive gap means the destination needs more DR bootstrap capacity.
A.8	Shared-DR overcommit ratio	`overcommit_ratio(dest) = sum_of_portions / max_portion` ; measures how oversubscribed the shared

Code	Name	One-line definition
		standby is; used in FIN-008 cost accounting.
A.9	Even zone-distribution target & skew	`target_per_zone = round(total / zone_count)`; `skew = max_count - min_count`; rebalance triggered when skew > C-13 tolerance. Implements PLC-011.

C — Configurable Items Register (Appendix C)

Primary source: Requirements Baseline v2.4 — §22 Configurable Items Register

These are items where the capability is required but the specific value or policy is not fixed in the baseline — they are driven by `PlacementPolicy` configuration.

Code	Name	Current direction	Status
C-1	DR bootstrap capacity	Lean “enough to bootstrap”; discussed 30–40% → 10–20% → ~5% → used-is-free model	Configurable — no fixed value
C-2	Production reservation buffer	`allocated + buffer`; ref impl buffer = 1 in dev	Configurable by product/region/env/SKU
C-3	Production quota buffer	Headroom above usage to support growth	Configurable — value TBD
C-4	Reconciliation frequency	Ref impl every 6 minutes	Configurable — prod value TBD (POC-010)
C-5	Failback model	Prefer ~1-year run; ~30-day failback alternative	Business decision (DEC-002)
C-6	Onboarding selection mode	Exact production region (default); geography (exception)	Configurable + exception policy (DEC-003)
C-7	Quota grouping model	One governed pool preferred	Configurable
C-8	Region catalogue & flags	Five geographies (US 3-region; Europe/Australia/Asia Pacific/Middle East 2-region); restricted/standard classes; `DR_NOT_OFFERED`	Configurable (REG-001)
C-9	Reservation over-allocation	Track allocated; allow over-association with alert	Configurable policy (CAP-018)
C-10	DR drill duration/rotation	Annual drill; role flip	Business decision (DEC-002)
C-11	DR sizing basis	**Max over non-concurrent sources** (DR-017); sum avail-	Configurable — max is default (Appendix D)

Code	Name	Current direction	Status
		able as conservative per-scope override	
C-12	Resource naming convention & counter	Deterministic RG/CRG/subscription naming with env/region/purpose tokens + zero-padded counter (OPS-006); CRG scope tokens `reg/az1/az2/az3` per CAP-023	Configurable
C-13	Zone-distribution target & tolerance	Even ≈ `1/zone_count` per zone (≈33% in three-zone regions) with configurable skew tolerance before rebalancing (PLC-011, A.9)	Configurable

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POC / DEP — Pending Decisions & Mandatory POCs

Primary source: Requirements Baseline v2.4 — §23 Pending Decisions & Mandatory POCs

POC items are Azure behaviours that **must be validated** before a production dependency can be declared. DEP items are external dependencies gating design decisions.

Code	Name	Required outcome
POC-001	Capacity reservation sharing quota behaviour	Confirm whether the consumer subscription must hold its own quota when consuming a shared reservation. **(Top technical unknown.)**
POC-002	Sharing consumption order	How Azure allocates shared capacity when multiple consumers request the same SKU simultaneously.
POC-003	Zone/subscription boundaries	Validate supported sharing combinations for the selected feature version (Preview or GA).
POC-004	Change under throttling	Safe reconciliation and retry behaviour during Azure API throttling.
POC-005	VM association/shutdown states	Reservation and guarantee behaviour across allocated/stopped/deallocated/associated/disassociated states.
POC-006	DR subscription topology	Dedicated DR subscription+cluster vs shared production subscription model for DR bootstrap.
POC-007	Bootstrap sizing	Minimum bootstrap per product including required control-plane nodes and SKUs.
POC-008	Production quota buffer	Initial buffer policies by product/VM family.
POC-009	Production reservation buffer	Initial reserved-capacity buffer by product/region/zone/SKU.
POC-010	Reconciliation interval	Production interval after API throttling and cost testing (C-4).

Code	Name	Required outcome
POC-011	Max-not-sum overcommit safety	Validate that shared/overcommitted DR reservations (DR-017) behave correctly when a standby set activates; quantify residual risk.
DEP-001	Azure feature maturity	Track Capacity Reservation Sharing preview → GA status and approved enterprise-usage conditions.

FR — Functional Requirements

Primary source: Complete Requirements Reference — Part 1

FR codes are the eight top-level functional capability areas; each expands to sub-requirements (FR-N.M).

Code	Name	One-line definition
FR-1	Capacity Reservation Lifecycle Management	Full CRUD lifecycle of CRGs and CRs across the Azure estate at API version 2024-03-01+.
FR-2	Capacity Reservation Sharing Management	Orchestrates the three-step RBAC setup for Provider-Consumer CRG sharing; manages subscription membership and 100-consumer limit.
FR-3	Availability Zone Mapping Management	Discovers, stores, and resolves logical-to-physical zone mappings across Provider-Consumer subscription pairs.
FR-4	Quota Management	Queries, tracks, and enforces quota constraints across managed subscriptions and quota groups.
FR-5	Disaster Recovery Failover Management	Manages pre-positioned DR capacity pairs, failover triggering, and failback orchestration.
FR-6	Regional Placement Decisions	Scores and selects Prod/CVAL/DR regions using placement scoring pipeline; creates and manages seed records.
FR-7	Capacity Forecasting	Forecasts demand growth and triggers proactive quota/reservation increase requests.
FR-8	Cost Optimization	Surfaces idle reservation cost, overcommit ratios, and FinOps metrics; supports configurable economic policy.

NFR-R — Placement Non-Functional Requirements

Primary source: Complete Requirements Reference — Part 3

Code	Name	One-line definition
NFR-R1	Placement API latency	Placement recommendation API P99 latency < 2 seconds for datasets with 100+ CRGs.
NFR-R2	Regional state freshness	Regional state used for placement must be ≤ 5 minutes old (via 5-min reconciliation or 10-min Cosmos DB fallback).
NFR-R3	Placement determinism	Placement scoring is deterministic and repeatable: same inputs always produce the same output; versioned policy enables replay.

R — Regional Placement Requirements

Primary source: Complete Requirements Reference — Part 3, Regional Placement Requirements

Code	Name	One-line definition
R1	Geography → Prod region derivation	Customer-supplied geography → engine derives Prod region via `argmax(PS_Prod)` over Standard regions in that geography.
R2	Automatic NonProd/DR selection	Engine selects NonProd and DR regions automatically; never same as Prod; allows NonProd=DR co-existence (PLC-010a).
R3	Hard constraints gate eligibility	Hard constraints HC-1..HC-10 gate all region eligibility before scoring.
R4	Automatic zone resolution	Automatic zone resolution via stored zone mapping registry on VM deployment against a shared CRG.
R5	Capacity-weighted distribution	Cost and capacity-weighted distribution prevent hotspots; uses demand units (vCPUs), not customer count.
R6	DR floor enforcement	NonProd placement is blocked if it would encroach on the `dr_floor_vcpu` of the target region (HC-7).
R7	Middle East special handling	`argmax(PS_Prod)` over Saudi Arabia Central + UAE North; cross-geo DR path conditional on DEC-001 clearance.
R8	Placement auditability	Placement is deterministic and auditable: all scores, candidate sets, and policy version are written to an OperationRecord for replay.

ADR — Architecture Decision Records

Primary source: Architecture/adr/

ADRs are formal, self-contained records of the load-bearing architectural decisions with Context / Decision / Consequences / Alternatives-considered sections. Each includes rendered architecture diagrams.

Code	Name	One-line summary	Full document
ADR-001	Region Selection & Placement	Geography-aware placement scoring pipeline (PS_Prod / PS_NonProd / PS_DR), HC-1..HC-10 filter pipeline, validation rules VR-1..VR-11, and the Customer-SeedRecord model.	acrme_adr_001_region_selection.md
ADR-002	Quota & Capacity Management	Single governed quota pool, two-group model (Provider + Consumer), DR-floor accounting formula, max-not-sum sizing, and quota increase workflow.	acrme_adr_002_quota_and_capacity_management.md
ADR-003	Capacity Management During DR	Distributed DR reference model, three-tier sharing model, DR activation sequence, CVAL earmark, engine state machine, and no-double-count rule.	acrme_adr_003_capacity_management_during_dr.md
ADR-004	Forecast & Increase of Capacity & Quota	10-step steady-state lifecycle, 5-phase auto-increase trigger model, forecast horizon, debounce/settle policy, and auto-decrease exclusion rule.	acrme_adr_004_forecast_and_increase_of_capacity_and_quota.md
ADR-005	Distributed DR Reference Model	Multi-source multi-destination DR topology, overcommit accounting, annual mock DR drill, failback orchestration, and the max-not-sum sizing correction.	acrme_adr_005_distributed_dr_reference_model.md

G — Design Gaps

Primary source: UML Class Diagrams Summary — Known Gaps

G-codes identify **explicitly documented open design items** — the “30% gap” declared in the UML class diagrams summary. These are design decisions, not research tasks.

Code	Name	One-line definition	Impact
G-14	Consumer credential model (Tier 3)	No decision yet on how ACRME authenticates to a consumer-owned resource group to forcibly disassociate a VM (customer-provisioned UAMI? Cross-tenant SPN? Delegated consent?).	Tier 3 VM disassociation cannot be implemented until resolved.
G-15 / B-3	Engine mode state machine incomplete	`engine_mode` transition guards, concurrency model, crash-recovery sequence, and operator API contracts are not yet fully specified.	Reconciliation loop and runner core cannot be production-hardened.
G-20	Cosmos DB entity schemas	Partition key strategy, secondary indexes, TTL policies, and consistency levels for all entities are not yet designed.	Data-layer implementation cannot begin; partition keys are irreversible after production data is written.
G-21	FastAPI service contracts	Request/response DTOs, RBAC enforcement middleware, API versioning, and endpoint specifications for all services are not yet defined.	Inter-service integration cannot be coded or tested.

ACRME-E — Backlog Epics

Primary source: Backlog Epics, Stories & Tasks — Epic Index

Each epic contains stories (`ACRME-S####`) and tasks (`ACRME-T######`). 20 epics / 72 stories / 193 tasks / 457 story points total.

Code	Epic name	Stories	Points	Phase
ACRME-E01	Foundation & Platform Infrastructure	5	39	P1
ACRME-E02	Subscription & Onboarding Lifecycle	4	34	P1
ACRME-E03	CRG & Capacity Reservation Management	3	21	P1
ACRME-E04	CRG Sharing & Consumer Authorization	4	21	P1
ACRME-E05	Quota Group Management	4	21	P1
ACRME-E06	Cross-Subscription Zone Alignment	2	13	P1
ACRME-E07	Region Selection & Placement Engine	8	49	P1
ACRME-E08	Placement Scoring & Forecasting	4	23	P1/P2
ACRME-E09	State Model & Concurrency Controls	2	21	P1
ACRME-E10	DR Activation & Failback	3	24	P1
ACRME-E11	Tier Escalation (Emergency Capacity)	3	19	P1
ACRME-E12	AKS & VMSS Integration	2	13	P2
ACRME-E13		2	13	P1

Code	Epic name	Stories	Points	Phase
	Data Architecture & Entity Model			
ACRME-E14	API Architecture	3	16	P1
ACRME-E15	Security, RBAC & Managed Identity	3	19	P1
ACRME-E16	Observability, Dashboards & Alerts	3	15	P1
ACRME-E17	Reconciliation & Scaling	4	23	P1
ACRME-E18	POC & Validation Program	4	29	P1
ACRME-E19	Production Readiness Gates & Governance	3	13	P1
ACRME-E20	v2.4 Reservation-Model Reconciliation	6	31	P1

Complete Code Inventory (Alphabetical)

Quick-scan index of every code in the corpus, in alphabetical/numerical order.

Code	Family	One-liner
A.1	Formulas	Reservation target: `reserved = allocated + buffer`
A.2	Formulas	Reservation headroom: `headroom = reserved - allocated`
A.3	Formulas	Reservation deficit: `deficit = allocated - reserved`
A.4	Formulas	Available subscription quota: `quota_limit − quota_current_value`
A.5	Formulas	Deployment quota deficit: `committed_vCPUs − available_quota`
A.6	Formulas	DR destination requirement (max, not sum)
A.7	Formulas	DR capacity gap: `DR_req − dr_reserved`
A.8	Formulas	Shared-DR overcommit ratio
A.9	Formulas	Even zone-distribution target and skew (PLC-011)
ACRME-E01..E20	Backlog	Foundation → Production Readiness (20 delivery epics)
ADR-001	Architecture	Region selection and placement decision record
ADR-002	Architecture	Quota and capacity management decision record
ADR-003	Architecture	Capacity management during DR decision record
ADR-004	Architecture	Forecast and increase of capacity decision record
ADR-005	Architecture	Distributed DR reference model decision record

Code	Family	One-liner
C-1	Configurable	DR bootstrap capacity
C-2	Configurable	Production reservation buffer
C-3	Configurable	Production quota buffer
C-4	Configurable	Reconciliation frequency
C-5	Configurable	Failback model
C-6	Configurable	Onboarding selection mode
C-7	Configurable	Quota grouping model
C-8	Configurable	Region catalogue and flags
C-9	Configurable	Reservation over-allocation policy
C-10	Configurable	DR drill duration/rotation
C-11	Configurable	DR sizing basis (max vs sum)
C-12	Configurable	Resource naming convention and counter
C-13	Configurable	Zone-distribution target and tolerance
CAP-001	Capacity	Authoritative managed scope
CAP-001a	Capacity	Core subscription = all production
CAP-002	Capacity	Azure resource must precede config activation
CAP-003	Capacity	Reservation target formula
CAP-004	Capacity	Associated-but-deallocated VMs
CAP-005	Capacity	Automated reconciliation
CAP-006	Capacity	Reconciliation frequency
CAP-007	Capacity	Scale-up behaviour

Code	Family	One-liner
CAP-008	Capacity	Scale-down behaviour
CAP-009	Capacity	Zero-capacity support
CAP-010	Capacity	No automatic deletion
CAP-011	Capacity	Availability-zone isolation
CAP-012	Capacity	Regional isolation
CAP-013	Capacity	Capacity sharing
CAP-014	Capacity	Shared capacity visibility
CAP-015	Capacity	No double counting
CAP-016	Capacity	Reservation integrity validation
CAP-017	Capacity	Deployment failure policy
CAP-018	Capacity	Over-allocation policy
CAP-019	Capacity	Scope-file governance
CAP-020	Capacity	Availability-Set VMs ineligible (v2.4)
CAP-021	Capacity	Deallocate-or-migrate-to-AZ precondition (v2.4)
CAP-022	Capacity	Seed-at-0 matrix and budget governance (v2.4)
CAP-023	Capacity	Regional and per-AZ CRG structure (v2.4)
CAP-024	Capacity	Reactive SKU/AZ discovery (v2.4)
DAT-001..006	Data	State store, entities, freshness, history, versioning
DEC-001	Decision	Middle East DR offering
DEC-002	Decision	DR drill duration and failback
DEC-003	Decision	

Code	Family	One-liner
		Geography exception approv-al
DEP-001	Dependency	Azure feature maturity (CR Sharing Preview→GA)
DR-001	DR	Single-region failure basis
DR-002	DR	Distributed DR model
DR-003	DR	Destination distribution
DR-004	DR	CVAL target contribution
DR-005	DR	CVAL may exceed DR need
DR-006	DR	Staged DR capacity acquisi-tion
DR-007	DR	DR target is configurable
DR-008	DR	Control plane first
DR-009	DR	Recovery prioritisation
DR-010	DR	DR declaration guardrail
DR-011	DR	Source region unavailable during outage
DR-012	DR	DR drill rotation
DR-013	DR	Failback policy
DR-014	DR	Middle East DR_NOT_OFFERED flag
DR-015	DR	Future active-active note
DR-016	DR	Reciprocal multi-source host-ing
DR-017	DR	Non-concurrent max-not-sum sizing
DR-018	DR	Source→destination DR index
DR-019	DR	

Code	Family	One-liner
		Standby activation on declaration
ENV-001..007	Environment	Production through R&D environment policy rules
FIN-001..008	FinOps	Idle cost, buffer, overcommit accounting
FR-1..8	Functional	Top-level functional capability areas
G-14	Design gap	Consumer credential model (Tier 3) unresolved
G-15/B-3	Design gap	Engine mode state machine incomplete
G-20	Design gap	Cosmos DB entity schemas not yet designed
G-21	Design gap	FastAPI service contracts not yet defined
GOV-001..009	Governance	Least privilege through data classification
HC-1..11	Hard constraints	Blocking placement rules (region, capacity, quota, zone)
INT-001..007	Integration	AEP contract, idempotency, input/output spec
NFR-001..010	Non-functional	Availability, consistency, performance, idempotency
NFR-R1..3	Placement NFR	Latency, freshness, determinism
OBS-001..005	Observability	Metrics, alerts, dashboards, SLO
OPS-001..006	Operational	Runbooks, pause, override, naming convention
PLC-001..011 + PLC-010a	Placement	Seed record, scoring, zone distribution

Code	Family	One-liner
POC-001..011	POC	Azure behaviour validation gates
QUA-001..014	Quota	Pooling, allocation, reclamation, governance
R1..8	Placement	Specific placement behaviours
RDY-001..004	Readiness	Deployment readiness gate and state model
REG-001..005	Region	Region catalogue, distribution model, cross-geo
VR-1..11 + VR-8a	Validation	Named enforcement checks for hard constraints

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This index is generated from the v2.4 baseline. When new codes are introduced, add them here first — this document is the master lookup point.